

**NOV-DEC 2025 EXAMINATION**  
**I B.TECH. (4YDC) EXAM**  
**MA10021: MATHEMATICS FOR ENGINEERS**

Time : 3 Hrs.]

[ Max. Marks :70

|                                                 |
|-------------------------------------------------|
| <b>TOTAL NO. OF QUESTIONS IN THIS PAPER : 5</b> |
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Note :Each question carry five subparts a, b, c, d and e. Attempt subparts a, b, c and any one from d or e in each question. All questions carry equal marks.

|     |                                                                                                                                                                            | Marks | CO  | BL   | PI    |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|------|-------|
| Q.1 | (a) What are the key contributions of Indian mathematician Ramanujan in the field of mathematics.                                                                          | 02    | CO1 | BL-1 | 1.1.1 |
|     | (b) If $u = x(1-y)$ , $v = xy$ then find the value of $\frac{\partial(u,v)}{\partial(x,y)}$ .                                                                              | 02    | CO1 | BL-2 | 1.1.1 |
|     | (c) If $u = \log(x^3 + y^3 + z^3 - 3xyz)$ then show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x+y+z}$ | 03    | CO1 | BL-2 | 1.1.1 |
|     | (d) Show that the rectangular solid of maximum volume that can be inscribed in a given sphere is a cube.                                                                   | 07    | CO1 | BL-3 | 2.1.3 |
|     | <b>OR</b>                                                                                                                                                                  |       |     |      |       |
|     | (e) Expand $\sin xy$ in powers of $(x-1)$ and $\left(y - \frac{\pi}{2}\right)$ up to second degree term.                                                                   | 07    | CO1 | BL-2 | 2.1.3 |
| Q.2 | (a) Evaluate $\int_0^1 x^5 (1-x^3)^{10} dx$ .                                                                                                                              | 02    | CO2 | BL-2 | 2.1.3 |
|     | (b) Determine the value of $\Gamma(-3/2)$ .                                                                                                                                | 02    | CO2 | BL-2 | 2.1.3 |
|     | (c) Show that $\int_0^{\pi/2} \frac{dx}{\sqrt{\sin x}} \times \int_0^{\pi/2} \sqrt{\sin x} dx = \pi$ .                                                                     | 03    | CO2 | BL-2 | 2.1.3 |
|     | (d) Change the order of integration in the integral $\int_0^{1-x} \int_{x^2}^{2-x} xy dy dx$ and evaluate it.                                                              | 07    | CO2 | BL-3 | 2.1.3 |
|     | <b>OR</b>                                                                                                                                                                  |       |     |      |       |
|     | (e) Evaluate the integral $\iint r^3 dr d\theta$ over the area bounded between the circles $r = 2\cos\theta$ and $r = 4\cos\theta$ .                                       | 07    | CO2 | BL-3 | 2.1.3 |
| Q.3 | (a) Determine the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}$                                                               | 02    | CO3 | BL-2 | 1.1.1 |
|     | (b) If $\lambda$ be an eigen values of a non-singular matrix $A$ , then prove that $\lambda^{-1}$ is an eigen values of $A^{-1}$ .                                         | 02    | CO3 | BL-2 | 1.1.1 |
|     | (c) Find the eigen values of the matrix $A = \begin{bmatrix} 3 & 2 \\ -1 & 0 \end{bmatrix}$                                                                                | 03    | CO3 | BL-2 | 1.1.1 |

- (d) Determine the values of  $a$  and  $b$  for which the system  $x + 2y + 3z = 6$ ,  $x + 3y + 5z = 9$ ,  $2x + 5y + az = b$  has (i) no solution (ii) unique solution (iii) infinite number of solution.

OR

- (e) Verify Cayley Hamilton theorem for the matrix  $\begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$ . Also find  $A^4$  with the help of Cayley Hamilton theorem.

- Q.4 (a) Define order and degree of differential equation with example. 02 CO4 BL-1 2.1.3  
 (b) Find the particular integral of  $(D^2 + 4)y = \cos 2x$ . 02 CO4 BL-2 2.1.3  
 (c) Find the complementary function of the differential equation  $x^2 \frac{d^2 y}{dx^2} + 4x \frac{dy}{dx} + 2y = 0$  03 CO4 BL-2 2.1.3  
 (d) Solve  $\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + 2y = e^x \tan x$  07 CO4 BL-2 2.1.3

OR

- (e) A particle of mass  $m$  is projected vertically upward under gravity, the resistance of the air being  $mk$  times the velocity. Show that the greatest height attained by the particle is  $\frac{V^2}{g} [\lambda - \log(1 + \lambda)]$ , where  $V$  is the greatest velocity which the above mass will attain when it falls freely and  $\lambda V$  is the initial velocity. 07 CO4 BL-3 2.4.1

- Q.5 (a) Write the real-world applications of fuzzy set theory. 02 CO5 BL-1 1.1.1  
 (b) Let  $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$  and  $\tilde{A} = \{(2, 0.4), (3, 0.6), (4, 0.8), (5, 1), (6, 0.8), (7, 0.6), (8, 0.4)\}$ . Find scalar and relative cardinality of the fuzzy set  $\tilde{A}$ . 02 CO5 BL-2 1.1.1  
 (c) Distinguish between classical set and fuzzy set. 03 CO5 BL-1 1.1.1  
 (d) Define membership function. Describe the representation of membership function and its types. 07 CO5 BL-1 1.1.1

OR

- (e) (i) Define convexity, support and normality of a fuzzy set. 07 CO5 BL-2 1.1.1  
 (ii) If fuzzy sets  $\tilde{A}$ ,  $\tilde{B}$  and  $\tilde{C}$  are defined as  $\tilde{A} = \{(0, 0), (1, 0.33), (2, 0.5), (3, 0.6), (4, 0.66), (5, 0.71)\}$ ,  $\tilde{B} = \{(0, 1), (1, 0.5), (2, 0.25), (3, 0.125), (4, 0.062), (5, 0.031)\}$ ,  $\tilde{C} = \{(0, 0.02), (1, 0.09), (2, 1), (3, 0.09), (4, 0.02), (5, 0.011)\}$ . Then find  $A \cup B$ ,  $B \cup C$ , complement of  $A \cup C$  and complement of  $B \cap C$ .

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